

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To obtain the classification of various classes using k-Nearest Neighbor approach	
Experiment No: 03	Date:	Enrollment No: 92301733023

Aim: To obtain the classification of various classes using k-Nearest Neighbor approach

IDE: Google Colab **Theory:**

This algorithm is used to solve the classification model problems. K-nearest neighbor or K-NN algorithm basically creates an imaginary boundary to classify the data. When new data points come in, the algorithm will try to predict that to the nearest of the boundary line. Therefore, larger k value means smother curves of separation resulting in less complex models. Whereas, smaller k value tends to over fit the data and resulting in complex models. It's very important to have the right k-value when analyzing the dataset to avoid over fitting and under fitting of the dataset.

The model representation for KNN is the entire training dataset. It is as simple as that. KNN has no model other than storing the entire dataset, so there is no learning required. Efficient implementations can store the data using complex data structures like k-d trees to make look-up and matching of new patterns during prediction efficient. Because the entire training dataset is stored, you may want to think carefully about the consistency of your training data. It might be a good idea to curate it, update it often as new data becomes available and remove erroneous and outlier data.

K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories. K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm. K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems.

K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data. It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset. KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

Example: Suppose, we have an image of a creature that looks similar to cat and dog, but we want to know either it is a cat or dog. So for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find the similar features of the new data set to the cats and dogs images and based on the most similar features it will put it in either cat or dog category.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset

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4. Pre-process the data
5. Visualize the Data
6. Separate the feature and prediction value columns
7. Select the number K of the neighbors
8. Calculate the Euclidean distance of K number of neighbors
9. Take the K nearest neighbors as per the calculated Euclidean distance.
10. Among these k neighbors, count the number of the data points in each category.
11. Assign the new data points to that category for which the number of the neighbor is maximum.
12. Our model is ready.

Pre Lab Exercise:

- a. What is K in KNN-approach?

K represents the number of nearest neighbors considered while making a prediction.

The algorithm looks at the K closest data points and assigns the class based on majority voting among them.

- b. Which are the types of distance metrics?

Common distance metrics used in KNN are:

1. **Euclidean Distance** (most common)
2. **Manhattan Distance**
3. **Minkowski Distance**
4. **Hamming Distance** (for categorical data)

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c. What can be the criteria of selection of the value of K?

1. Choose K based on dataset size
2. Use **odd values** (to avoid ties)
3. Use **cross-validation** to find best K
4. Small K → Overfitting
5. Large K → Underfitting

d. What are the advantages of KNN algorithm?

1. Simple and easy to implement
 2. No training phase required
 3. Works well with small datasets
 4. Can be used for both classification and regression
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Program (Code):

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score
```

```
data = {
    'Study_Hours': [2,5,3,6,4,7,8,1,9,5],
    'Marks': [55,78,50,88,70,90,95,40,96,75],
    'Result': ['Fail','Pass','Fail','Pass','Pass','Pass','Pass','Fail','Pass','Pass']
}
```

```
df = pd.DataFrame(data)
```

```
X = df[['Study_Hours', 'Marks']]
y = df['Result']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
k = 3
model = KNeighborsClassifier(n_neighbors=k)
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)
```

```
print("Confusion Matrix:\n", cm)
```

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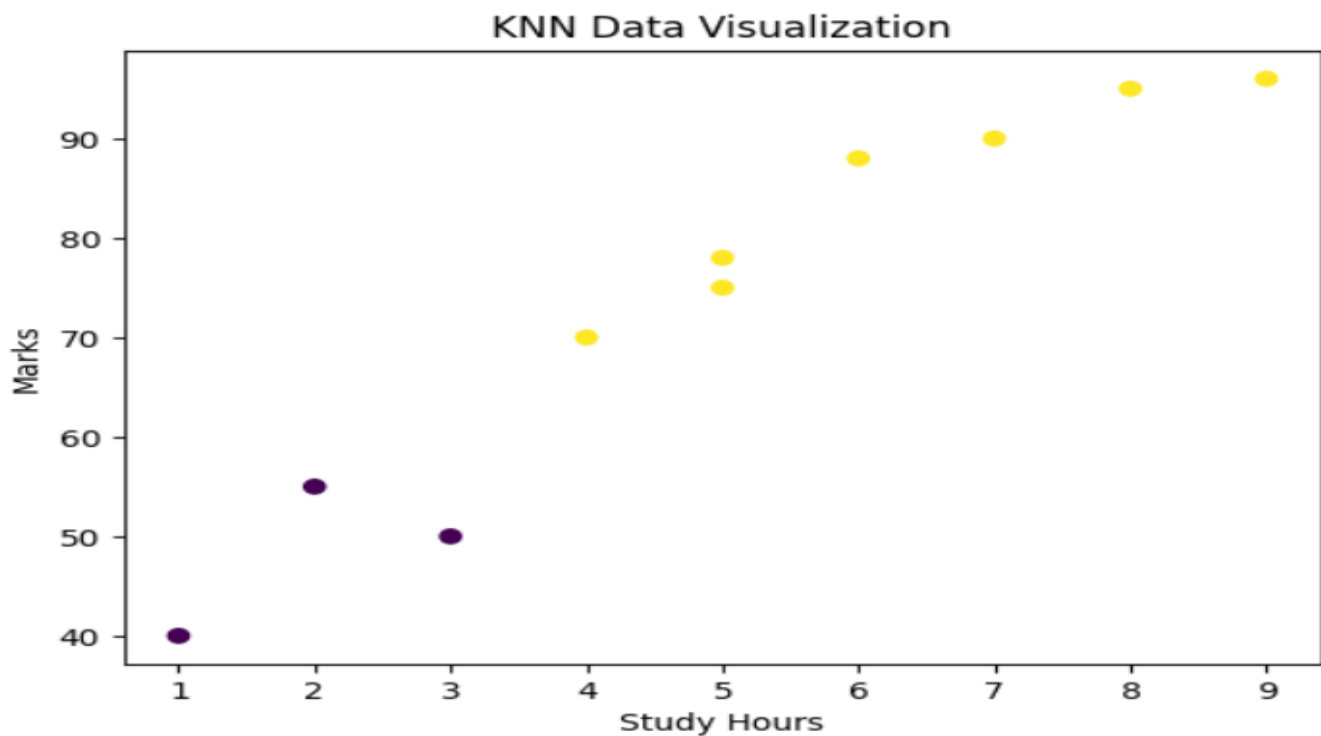
```
print("Accuracy:", acc)
```

```
plt.scatter(df['Study_Hours'], df['Marks'], c=(df['Result'] == 'Pass'))
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("KNN Data Visualization")
plt.show()
```

Results:

Confusion Matrix: [[2]]

Accuracy: 1.0



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Observation and Result Analysis:

a. Nature of the dataset

The dataset contains numerical features such as **Study Hours and Marks**, and a categorical target variable (**Result: Pass/Fail**).

It shows a clear pattern where higher study hours and marks correspond to the "Pass" category, indicating a **classifiable dataset** suitable for KNN.

b. During Training Process

1. KNN does not build an explicit model; it simply stores the training data.
2. The algorithm calculates the **distance (Euclidean)** between points.
3. The value of **K = 3** determines how many neighbors are considered.
4. No parameter learning occurs, making it a **lazy learning algorithm**.

c. After the training Process

- The model predicts test data based on nearest neighbors.
- Classification is done by **majority voting**.
- Confusion matrix shows correct and incorrect predictions.
- Accuracy score indicates model performance.

d. Observation over the Learning Curve

- KNN does not have a traditional learning curve since it does not train iteratively.
- Performance depends on value of K:
- Small K → Overfitting
- Large K → Underfitting
- Optimal K gives best accuracy.

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Post Lab Exercise:

a. Is KNN useful in regression-based problems?

Yes, KNN can be used for regression by taking the average of nearest neighbors, but it is mostly used for classification.

b. Why KNN is a non-parametric algorithm?

KNN does not assume any underlying data distribution and does not learn parameters; it directly uses the dataset for prediction.

c. What are the assumptions of KNN approach?

1. Similar data points exist close to each other
2. ~~Data is properly scaled~~
3. Distance metric reflects similarity

d. How can the KNN approach make the prediction of the unseen dataset?

1. Calculate distance from new point to all training points
2. Select K nearest neighbors
3. Use majority voting to assign class

e. Why is KNN called a Lazy Learner?

Because it does not learn during training; it only stores data and performs computation during prediction.

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f. Why should we not use KNN for large dataset?

1. High computation cost
2. Slow prediction
3. Requires large memory
